

IoT Based Tunnel Monitoring System

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Problem Statement

Train tunnels are critical infrastructures that require constant monitoring to ensure the safety of passengers and the efficient operation of trains. These underground environments are susceptible to various hazards, such as **temperature fluctuations, gas leaks, structural vibrations, and water ingress**, all of which can lead to severe accidents if not detected in time. Traditional methods of tunnel inspection are often time-consuming, labour-intensive, and reactive, addressing issues only after they occur. By implementing an IoT-based train tunnel monitoring system, **real-time** sensor data can be continuously collected and analysed. This proactive approach enables **early detection of potential threats, timely maintenance, and enhances overall tunnel safety**. Such a system prevents accidents and ensures the seamless operation of railway services, reducing downtime and improving passenger confidence.

Objective

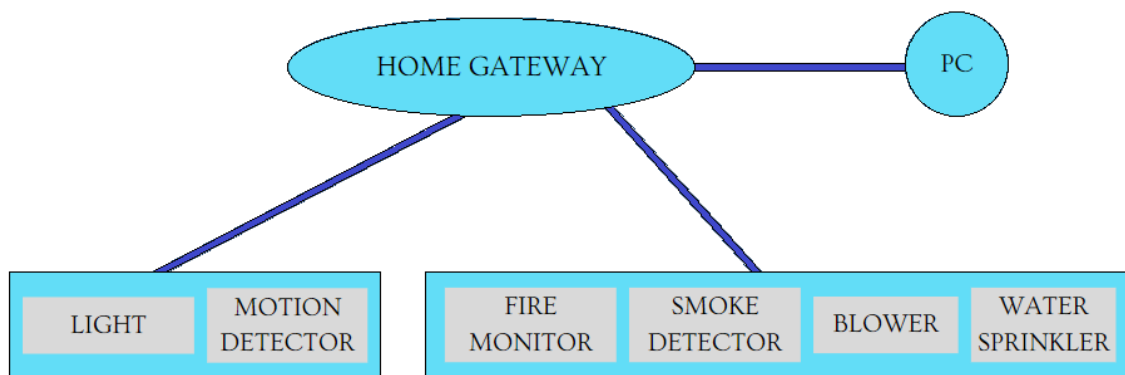
To design & implement an IoT system for **real-time monitoring** of **train tunnel** environments using various sensors connected to a home gateway, ensuring tunnel safety and operational efficiency.

Methodology

1. Design Network Topology (**Mesh**)
2. IP address Planning
3. Import Necessary Components
4. Configure Device Properties
5. Assign IP Address
6. Setting Up Server
7. IoT Gateway Configuration
8. Network Simulation
9. Test Automation Rules (Automation Logic)
10. Network Troubleshooting

Components and Connections

For Motion Detection	For Fire and Smoke Detection
Home Gateway	PC
PC	Fire Monitor
Motion Detector	Smoke Detector
Light	Blower
	Water Sprinkler



Real-Time Application

Real-Time Tunnel Monitoring System for early detection of faults and hazards.

For safe exit of the passengers from the tunnel incase the train stops working inside the tunnel

For maintenance of the tunnel

Result

